



Banking Analytics Dashboard

An end-to-end data analytics project combining **Python-based EDA** with **Power BI dashboarding** to transform raw banking data into actionable business insights.

PYTHON

POWER BI

BANKING ANALYTICS



Business Objective

The banking industry generates vast volumes of transactional and customer data. Without proper analysis, identifying trends and monitoring performance becomes difficult. This project addresses these challenges:



Customer Behavior

Analyze financial behavior across customer segments



Performance Monitoring

Track lending and deposit portfolio performance



KPI Development

Create business-focused key performance indicators



Strategic Decisions

Support data-driven decision-making

Dataset Overview

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3,000 Customers

Age, Gender, Nationality,
Occupation

Lending Products

Bank Loans, Business Lending,
Credit Card Balance

Deposit Products

Bank Deposits, Savings, Checking,
Foreign Currency

Relationship Types

Retail, Institutional, Private Bank,
Commercial

Exploratory Data Analysis — Python

01

Data Loading

Imported dataset using `pandas`; explored structure with `df.info()` and `df.describe()`

03

Feature Engineering

Categorized customers into Income Bands: **Low, Med, High**

02

Cleaning & Preparation

Standardized column names to `snake_case`; validated financial balances; no negative values where inappropriate

04

Correlation Analysis

Savings & Checking Accounts positively contribute to total deposits; Foreign Currency Accounts add customer value; Lending products correlate with income and deposit behavior

EDA — Key Insights



Customer Segmentation

Customers distributed across multiple relationship types, enabling targeted product offerings and personalized services



Deposit Portfolio

Deposits are not uniformly distributed — clear high-value and low-value deposit segments exist



Loan Portfolio

Loan balances vary significantly across customers, reflecting diverse borrowing patterns and financial profiles

KPI Development

Key Performance Indicators built using **DAX in Power BI** to evaluate lending, deposits, and customer value.

3,000

Total Customers

Unique customers in dataset

4.38B

Total Loans

Bank Loans + Business Lending + Credit Cards

3.77B

Total Deposits

Savings, Checking, Foreign Currency + Bank Deposits

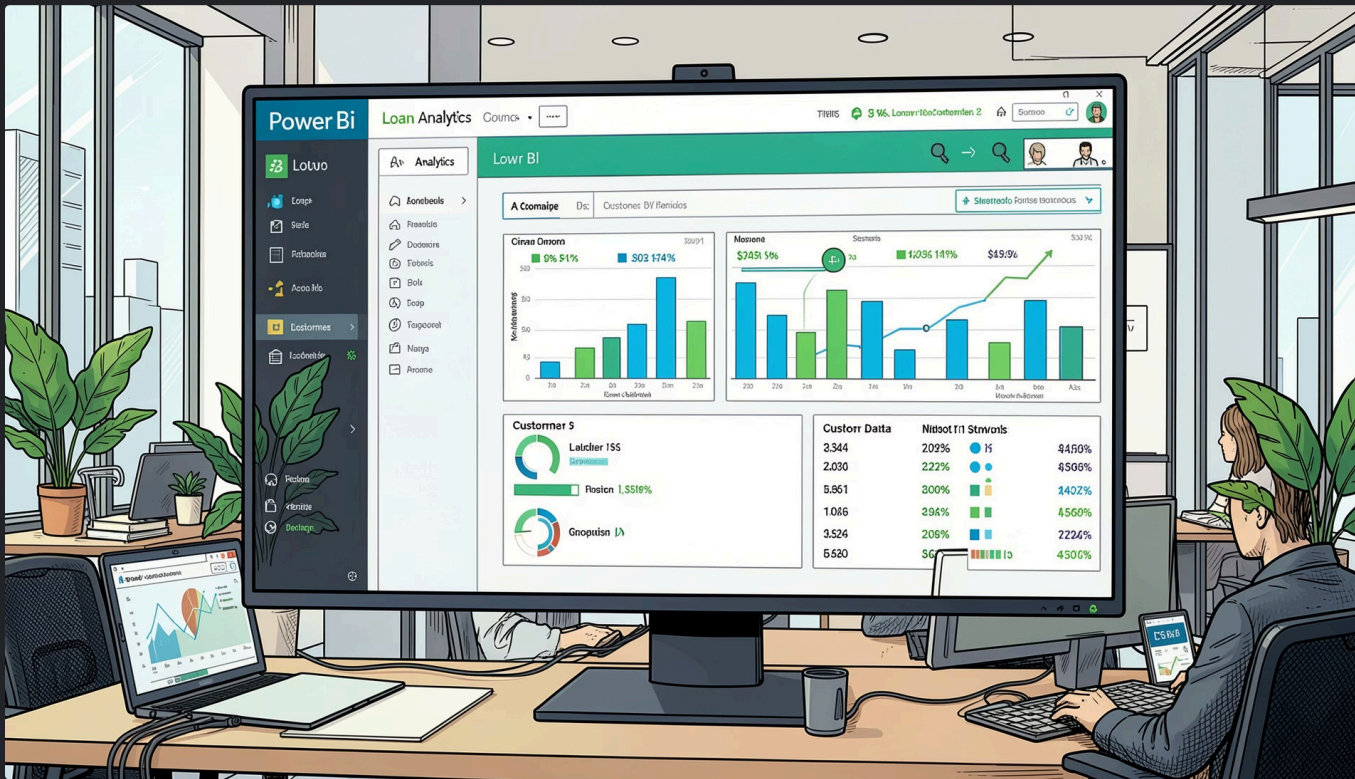
116%

Loan-to-Deposit Ratio

Bank lending exceeds deposits — liquidity management needed

 Avg Loan per Customer: **1.46M** | Avg Deposit per Customer: **1.28M**

Dashboard — Home & Loan Analysis



Home Page

High-level summary of total customers and core financial balances with slicers for Year, Gender, and Banking Relationship.

Loan Analysis Page

Detailed lending breakdown by demographics, occupation, income band, and nationality. Business Lending leads at **2.60B** vs Bank Loans at **1.77B**. High-income band holds **53.12%** of bank loans.

Dashboard — Deposit & Summary

Deposit Analysis

Visualizes savings vs. checking trends and income-based deposit clusters. High-income customers hold **54.17%** of bank deposits. Private Bank leads relationship types at **0.93B**. European nationality dominates at **0.87B**.

Summary Dashboard

Advanced metrics including LDR, average balances per customer, and foreign currency balance (**89.65M**).



Business Recommendations

1

Grow Customer Deposits

Introduce attractive savings products and loyalty programs to improve liquidity

2

Strengthen High-Value Relationships

Private Banking customers drive significant value — invest in personalized wealth management

3

Increase Credit Card Adoption

Credit cards are a small portion of lending — targeted promotions can boost uptake

4

Occupation-Based Solutions

Design customized loan and deposit products for high-engagement occupations

5

Personalized Marketing

Leverage demographic and relationship data for targeted campaigns and product recommendations

Conclusion & Tools

Project Journey

This project demonstrates the full analytics lifecycle – from data preparation and EDA to KPI creation and interactive dashboard development. The insights support smarter decisions in customer behavior, lending performance, and deposit management.

Tools & Technologies

Python

Pandas, NumPy

Visualization

Matplotlib, Seaborn

Power BI

DAX, Data Modeling

